Mengting **Li**

PhD Student in Bioinformatics

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Education

The University of Hong Kong (HKU) — *Hong Kong, China*

PhD in Bioinformatics — *Oct 2025 - Present*

University of Chinese Academy of Sciences (UCAS) — *Beijing, China*

Master of Biological and Pharmaceutical Engineering — *Sep 2020 - Jun 2023*

Xuzhou Medical University (XZMU) — *Xuzhou, China*

Bachelor of Bioinformatics — *Sep 2016 - Jun 2020*

Research Experience

Research Assistant (Jul 2023 - Sep 2025)
The University of Hong Kong | Hong Kong, China (Supervisor: Prof. Leo Poon)
Project: An interferon-stimulated long non-coding RNA USP30-AS1 as an immune modulator in influenza A virus infection

- Developed a reproducible bioinformatics pipeline that successfully processed and analyzed 5 RNA-seq datasets and 4 microarray datasets from public repositories (SRA, GEO).
- Identified 1,715 differentially expressed lncRNAs across multiple post-infection time points by implementing a custom differential expression analysis workflow (using `STAR`, `featureCounts`, and `DESeq2`).
- Revealed key immune pathways using `g:Profiler` for functional enrichment analysis and `EnrichmentMap` in `Cytoscape` for network visualization.

Master's Student Researcher (Sep 2020 - Jun 2023)
BGI-Research & University of Chinese Academy of Sciences | Shenzhen, China (Supervisor: Prof. Kui Wu)
Project: Multi-omics sequencing of gastroesophageal junction adenocarcinoma reveals prognosis-relevant key factors and a novel immunogenomic classification

- Led a multi-omics analysis of 92 gastroesophageal junction adenocarcinoma (GEJAC) patients, integrating whole-genomic, transcriptomic, and T-cell receptor repertoires data.
- Developed a comprehensive bioinformatics pipeline for processing WGS data to perform somatic variant calling and driver gene identification. Further utilized these results for downstream analyses including mutational signature analysis (`SigProfilerExtraction`), neoantigen prediction (`NetMHC`, `NetMHCpan`), and tumor evolution (`PhylogicNDT`).
- Discovered 30 mutated driver genes, 7 novel genomic signatures, 3 copy-number variations, and 2 V-J gene usages previously unidentified and linked to prognosis.
- Developed a novel immunogenomic classification system that stratifies patients into three immune-related subtypes (immune-inflamed, intermediate, and deserted) with discrete profiles of genomic signatures, immune status, and clinical outcomes.

Skills

Programming & Scripting Languages

- **R**: (Proficient) Data manipulation (`Tidyverse`, `data.table`), statistical modeling, bioinformatics (`Bioconductor`), visualization (`ggplot2`).
- **Python**: (Proficient) Scientific computing (`Pandas`, `NumPy`), machine learning (`PyTorch`, `Scikit-learn`), scripting and workflow integration.
- **Bash/Shell Scripting**: Proficient in automating tasks, managing data, and working in a Linux/HPC environment.

Bioinformatics & Genomics

- **Omics Analysis & Pipeline Development**: Proficient in building and deploying automated analysis pipelines for diverse genomic data modalities, including Whole Genome Sequencing (WGS), Bulk RNA-seq, single-cell RNA-seq (scRNA-seq), and Spatial Transcriptomics.
- **Phylogenetics & Evolutionary Analysis**: Experience in Bayesian phylogenetic inference (`BEAST`), maximum likelihood methods (e.g., `IQ-TREE`), and population genetics simulation (`SLiM`).
- **Sequence Analysis**: Viral lineage assignment (`Pangolin`), sequence alignment (`BWA`, `Bowtie2`), variant calling & manipulation (`VCFtools`, `BCFtools`), and general sequence analysis (`Samtools`, `BLAST`).
- **Genomic Data Formats**: Deep familiarity with standard formats including VCF, BAM/SAM, FASTQ, FASTA, and GFF/GTF.

Developer Tools & Workflow Management

- **Version Control**: `Git` / `GitHub` for collaborative and reproducible research.
- **Environment Management**: `Conda` for managing software dependencies and creating reproducible environments.
- **High-Performance Computing (HPC)**: Experience with job scheduling systems (e.g., `Slurm`, `SGE`) and working in a cluster environment.
- **Workflow Languages**: Familiarity with the principles of workflow management using `Snakemake` / `Nextflow`.

Deep Learning

- **Machine Learning Foundations**: Strong foundation in Regression, Classification, Clustering, DNNs, and CNNs.
- **Advanced Sequence Models**: Understanding of advanced architectures including RNNs, LSTMs, and Transformers.
- **Technical Implementation**: Hands-on project experience applying `Scikit-learn`, `PyTorch`, and `TensorFlow` to bioinformatics tasks.

Publications

1. Gu, H., Li, J., Sun, W., **Li, M.**, Leung, K., Wu, J. T., Yuan, H.-Y., Wang, M. H., Yang, B., McKay, M. R., Ning, N., & Poon, L. L. M. (2026). Optimizing global genomic surveillance for early detection of emerging SARS-CoV-2 variants. *Nature Communications*, 17, 4322. `https://doi.org/10.1038/s41467-026-70664-0`

2. Gu, H., **Li, M.**, Wang, R., Lee, L. K., Tam, W. Y., Ho, A. Y. M., Yau, M. C. Y., Yip, K. T., Wong, R. C. W., Wong, B. K. C., Luk, K. S., Lam, J. Y. W., Que, T. L., Chow, V. C. Y., Chau, S. K. Y., Duchene, S., Siu, G. K. H., & Poon, L. L. M. (2026). Genomic epidemiology of hospital-associated SARS-CoV-2 clusters in Hong Kong during a period of relaxed visitation (May–August 2022). *Influenza and Other Respiratory Viruses*, 20(3), e70249. `https://doi.org/10.1111/irv.70249`

3. Cao, Y., Chin, A. W. H., Gu, H., **Li, M.**, & Poon, L. L. M. (2025). Long non-coding RNA BCAR4 is required for efficient influenza A virus replication. *bioRxiv* [Preprint]. `https://doi.org/10.1101/2025.02.06.636885`

4. Ma, Z., **Li, M.**, Li, F., Wu, K., Wu, X., Luo, T., Gao, N., Luo, H., Sui, Z., Yu, Z., Jiang, H., Shang, X., Chen, C., Yue, J., Meng, F., Duan, X., & Xu, B. (2025). Multi-omics sequencing of gastroesophageal junction adenocarcinoma reveals prognosis-relevant key factors and a novel immunogenomic classification. *Gastric Cancer*, 28(3), 344–357. `https://doi.org/10.1007/s10120-025-01585-y`

5. Cao, Y., Chin, A. W. H., Gu, H., **Li, M.**, Gu, Y., Lau, S. P. N., Hui, K. P. Y., Chan, M. C. W., & Poon, L. L. M. (2025). An interferon-stimulated long non-coding RNA USP30-AS1 as an immune modulator in influenza A virus infection. *PLOS Pathogens*, 21(1), e1012854. `https://doi.org/10.1371/journal.ppat.1012854`

6. Zhang, J., & **Li, M.** (2020). Bioinformatics analysis and design based on RStudio and databases. *Modern Information Technology*, 4(4), 76–79. `https://qikan.cqvip.com/Qikan/Article/Detail?id=7101730810`